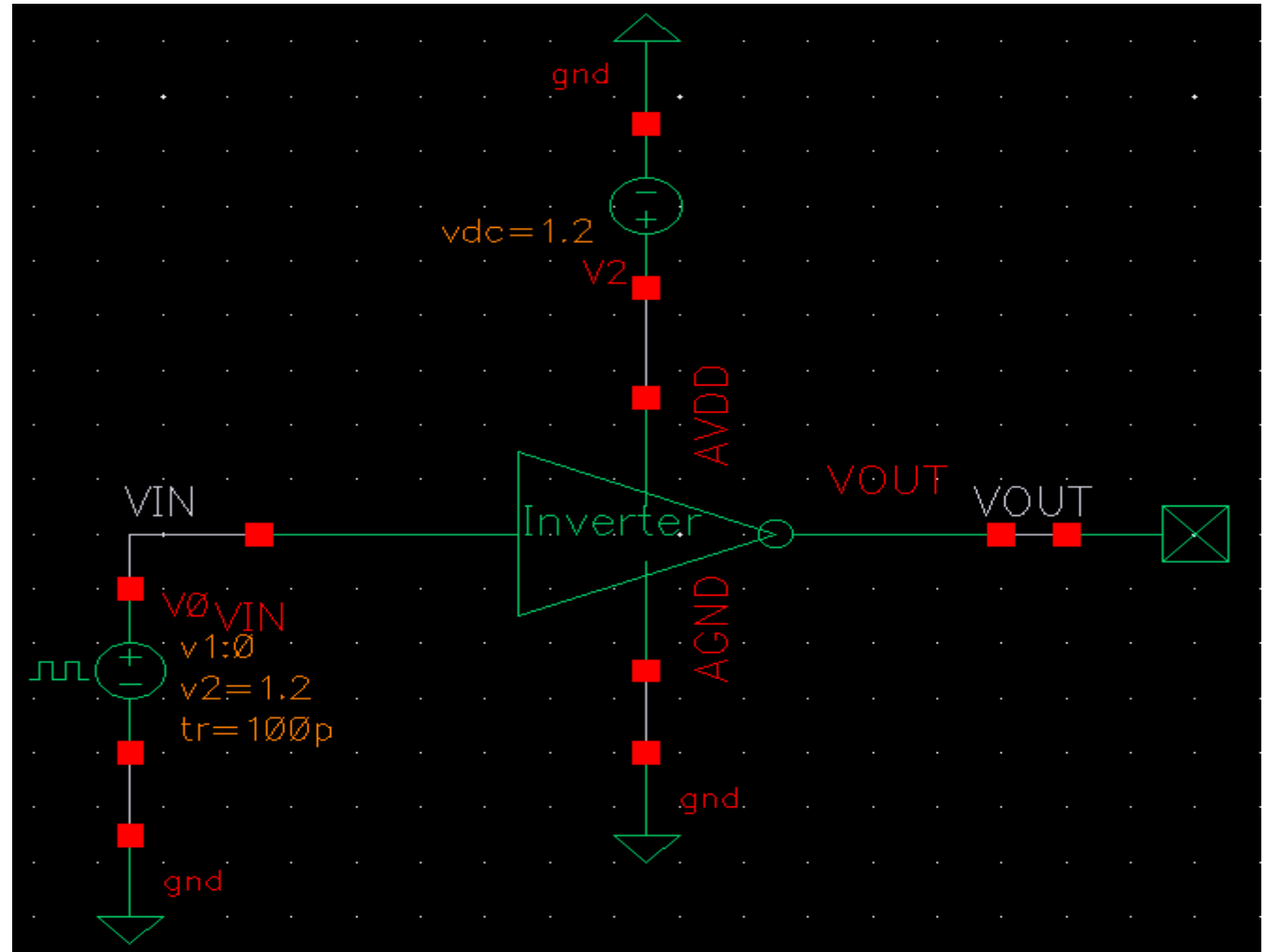
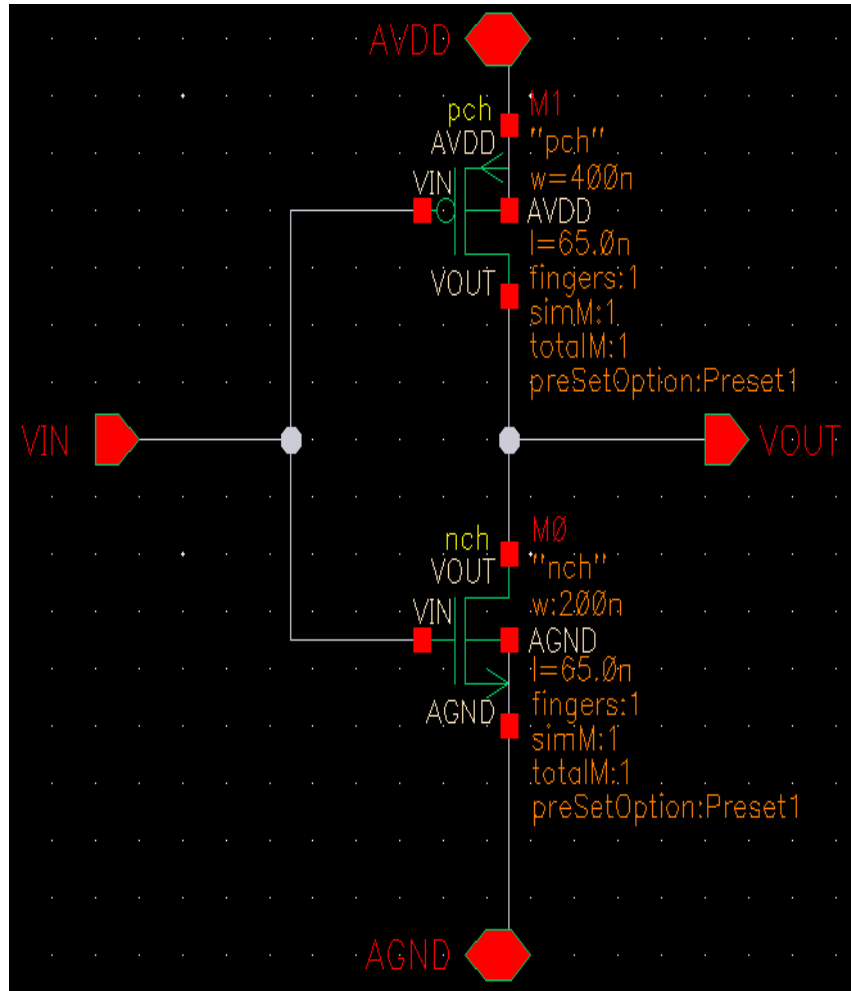
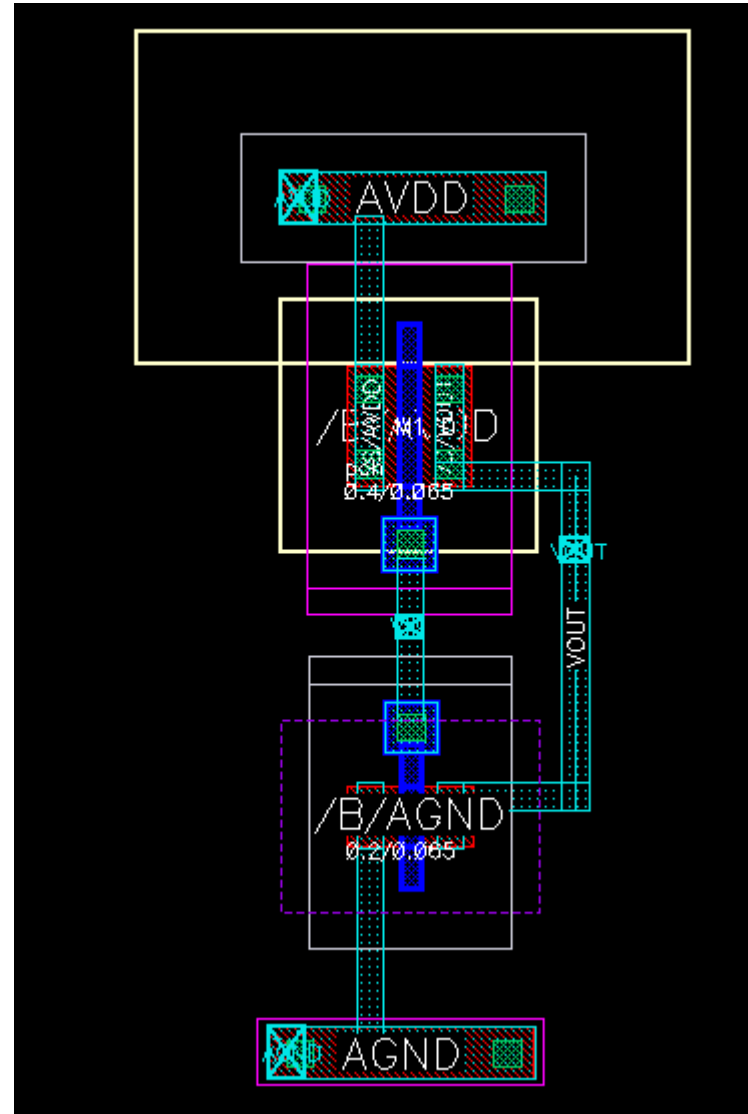


**Inverter Layout
using
TSMC 65nm Technology**

Inverter Schematic and Test Bench



Inverter Layout

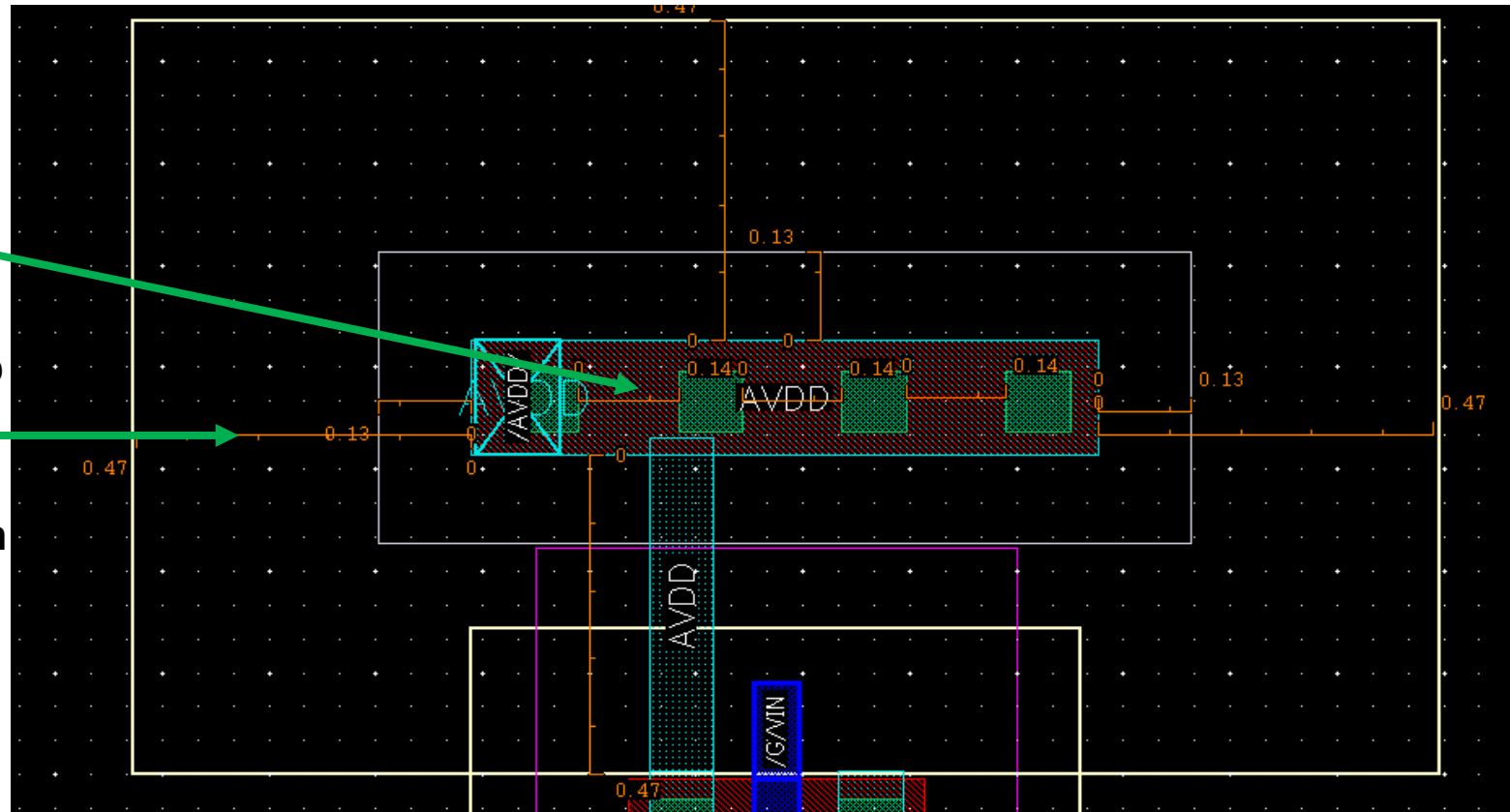


Inverter Layout Notes (Cont.)

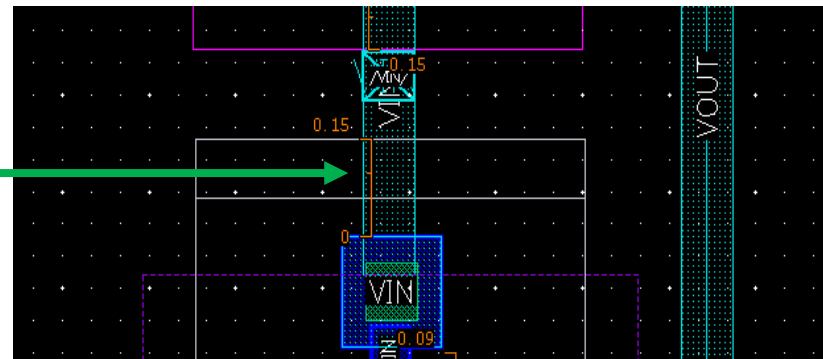
- Spacing between contacts and each other 0.14 μm

- Spacing between np and OD is 0.13 μm

- Spacing between OD and Nwell 0.47 μm



- Spacing between Poly and NP is 0.15 μm



DRC and LVS

Calibre - RVE v2014.1_17.12 : Inverter.drc.results

File View Highlight Tools Window Setup Help

Search

Show All Inverter, 0 Results (in 0 of 3 Checks)

Check / Cell	Results
Check DENSITY_PRINT_FILES	0

OD.DN.2L.density
OD.DN.2H_I0.density
OD.DN.2H_CORE.density
OD.DN.3L.density
OD.DN.3H_I0.density
OD.DN.3H_CORE.density
PO.DN.2.density
PO.DN.3.density

Check DENSITY_PRINT_FILES

Navigator

Comparison Results x ERC Results

Results

- Extraction Results
- Comparison Results

ERC

- ERC Results
- ERC Summary

Reports

- Extraction Report
- LVS Report

Rules

- Rules File

View

- Info
- Finder
- Schematics

Setup

- Options

Layout Cell / Type	Source Cell	Nets	Instances
Inverter	Inverter	4L, 4S	1L, 1S

Cell Inverter Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

```

#           #
#         #
#       #
#     #
#   #
# #
#
#####
#   CORRECT   #
#             #
#             #
#####

```

LAYOUT CELL NAME: Inverter
SOURCE CELL NAME: Inverter

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	4	4	
Nets:	4	4	
Instances:	1	1	MN (4 pins)
	1	1	MP (4 pins)
Total Inst:	2	2	

NUMBERS OF OBJECTS AFTER TRANSFORMATION

PEX and Result Viewing Environment (RVE)

Rules

Inputs

Outputs

Run Control

Transcript

Run PEX

Start RVE

```
--- NETWORK REDUCTION BEGIN:
--- READING FROM PDB...
--- BEGIN REDUCING NETS...
--- DONE REDUCING NETS...
--- WRITING TO PDB...
--- NETWORK REDUCTION COMPLETE: CPU TIME = 0  REAL TIME = 0  LVHEAP = 329/331/331  MALLOC = 339/339/339

-----
PDB NET SUMMARY
-----

pdb file name =      svdb/INVERTER.pdb
root cell name =      INVERTER
total nets =         4
top-level nets =      4
non-top-level nets =  0
degenerate nets =     0
merged nets =         0
error nets =          0

=====
CALIBRE xRC WARNING / ERROR Summary
=====

xRC Warnings = 1
xRC Errors = 0

=====

--- CALIBRE xRC: FORMATTER COMPLETED - Mon May 11 10:26:12 2026
--- TOTAL CPU TIME = 0  REAL TIME = 0  LVHEAP = 26/55/331  MALLOC = 307/307/339  ELAPSED TIME = 2
```

8 Warnings

Layer 6 contains unmapped objects and is the source layer of LAYER MAP == 6 DATATYPE == 1.
Layer 6 contains unmapped objects and is the source layer of LAYER MAP == 6 DATATYPE == 3.
Layer 17 contains unmapped objects and is the source layer of LAYER MAP == 17 DATATYPE == 1.
Layer 17 contains unmapped objects and is the source layer of LAYER MAP == 17 DATATYPE == 51.
Layer 31 contains unmapped objects and is the source layer of LAYER MAP == 31 DATATYPE == 1.
Contact layer nplug cannot appear as one of the connected layers on CONNECT statement. CONNECT statement ignored.
Contact layer pplug cannot appear as one of the connected layers on CONNECT statement. CONNECT statement ignored.
PEX IDEAL XCELL is only applied in gate-level extraction.

Calibre Info

Calibre View generation completed with 0 WARNINGS and 0 ERRORS.
Please consult the CIW transcript for messages.

Close

PEX Netlist File - Inverter.pex.netlist

File Edit Options Windows

; DESIGN "INVERTER"
; DATE "Mon May 11 10:26:12 2026"
; VENDOR "Mentor Graphics Corp."
; PROGRAM "Calibre xRC v2014.1_17.12"
; CIRCUIT TEMPERATURE 27C
; NOMINAL TEMPERATURE 27C
;
mgc_rve_device_template "pch_hv25_spw" "D" "G" "S" "B"
mgc_rve_device_template "pch_hv25_spw_mac" "D" "G" "S" "B"
mgc_rve_device_template "nch_hv25_snw" "D" "G" "S" "B"
mgc_rve_device_template "nch_hv25_snw_mac" "D" "G" "S" "B"
mgc_rve_device_template "nch" "D" "G" "S" "B"
mgc_rve_device_template "nch_18" "D" "G" "S" "B"
mgc_rve_device_template "nch_18_mac" "D" "G" "S" "B"
mgc_rve_device_template "nch_25" "D" "G" "S" "B"
mgc_rve_device_template "nch_25_mac" "D" "G" "S" "B"
mgc_rve_device_template "nch_33" "D" "G" "S" "B"
mgc_rve_device_template "nch_33_mac" "D" "G" "S" "B"
mgc_rve_device_template "nch_25od28" "D" "G" "S" "B"
mgc_rve_device_template "nch_25od28_mac" "D" "G" "S" "B"
mgc_rve_device_template "nch_ltr" "D" "G" "S" "B"
mgc_rve_device_template "nch_mac" "D" "G" "S" "B"
mgc_rve_device_template "nch_w" "D" "G" "S" "B"
mgc_rve_device_template "nch_w_lvt" "D" "G" "S" "B"
mgc_rve_device_template "nch_esd18" "D" "G" "S" "B"
mgc_rve_device_template "nch_esd18_mac" "D" "G" "S" "B"
mgc_rve_device_template "nch_hvt" "D" "G" "S" "B"
mgc_rve_device_template "nch_hvt_mac" "D" "G" "S" "B"

Edit Row 1 Col 1

Calibre - RVE v2014.1_17.12 : svdb Inverter

File View Highlight Tools Window Setup Help

Navigator

Results

Extraction Results

Comparison Results

Parasitics

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	AVDD	AVDD	11	1.03128E-16	9.11093E-17	1.94237E-16
2	AGND	AGND	11	1.14614E-16	5.32581E-17	1.67872E-16
3	VIN	VIN	10	8.41415E-17	1.74621E-16	2.58763E-16
4	VOUT	VOUT	13	1.14089E-16	1.65707E-16	2.79796E-16

Simulation Pre-Layout and Post-Layout results

